

Project 1: Scattering an electron off a potential

In this project, you will solve the classic problem of a non-relativistic quantum-mechanical particle (e.g., an electron) scattering off a potential barrier. We know from quantum mechanics that particles behave like both waves and particles. We can describe a quantum-mechanical particle using a wavefunction. A non-relativistic particle's wavefunction $\psi(x,t)$ satisfies the time-dependent Schrödinger equation

$$i\hbar \frac{\partial}{\partial t} \psi(x,t) = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x,t) \right] \psi(x,t)$$

where m is the mass of the particle and $V(x,t)$ is the potential energy of the particle. The Schrödinger equation is identical to the heat equation that we solved in Activity Guide 7, except that some of its coefficients are complex-valued. The result is that the wavefunction that solves this equation may be complex. We can use the wavefunction to find the probability of observing the particle at position x at time t via $p(x,t)=|\psi(x,t)|^2$, from which we can then calculate various observables such as average position, velocity, and energy.

In your Introduction, give a brief intro to the Schrödinger equation, for example what the terms represent.

Crank-Nicolson Method

Numerically, the same discretization procedure you used in solving the heat equation can be used on the time-dependent Schrödinger equation. In this project, we will look at a better numerical method for solving this equation.

Our previous approach for solving the heat/diffusion equation was an explicit method. We were able to get an explicit update rule for our function at time $t+\Delta t$. Here, we will use the Crank-Nicolson (CN) scheme, an *implicit* method because we end up having to solve a system of equations to find our function at the next time step. This method has the advantage of being super stable (i.e., has good convergence properties).

As in Activity Guide 7 on partial differential equations, we start by discretizing space into a 1D lattice of $(J+1)$ sites separated by Δx and discretizing time into steps of size Δt . Thus our wave function at position $x=x_j$ and at time $t=n\Delta t$ is denoted $\psi(x,t)=\psi_j^n$. Look up the Crank-Nicolson method and see if you can work out the following system of equations for finding the updated values of ψ_j^{n+1}

$$\begin{aligned} -\frac{1}{2}i\alpha\psi_{j-1}^{n+1} + \left[1 + i\alpha + \frac{i}{2}\frac{\Delta t}{\hbar}V(x_j) \right] \psi_j^{n+1} - \frac{1}{2}i\alpha\psi_{j+1}^{n+1} \\ = \frac{1}{2}i\alpha\psi_{j-1}^n + \left[1 - i\alpha - \frac{i}{2}\frac{\Delta t}{\hbar}V(x_j) \right] \psi_j^n + \frac{1}{2}i\alpha\psi_{j+1}^n \end{aligned}$$

where $\alpha=\hbar\Delta t/(2m\Delta x^2)$. The lefthand side involves the wave function at time $t+\Delta t$, whereas the righthand side only involves the previous time t . There are $J+1$ equations, giving a system of equations to solve for the unknown $\psi_{j,n+1}$ at each lattice site j . It's a triadiagonal system, just like the resistor network you solved back in Activity Guide 6 on linear algebra. So you should use 'solve_banded' from `scipy.linalg` to solve it really fast. We'll use boundary conditions that the wave function is zero on the boundaries, so the first and last equations in the system of equations are $\psi_{0,n+1}=\psi_{0,n}$ and $\psi_{J,n+1}=\psi_{J,n}$. You will have to set your arrays that you use in solving this system of equations to be of `dtype=complex`. The resulting solution will also be complex, having both imaginary and real parts.

In your Methods section, try to also state why the CN method is so stable compared to our previous explicit method.

Wavepackets

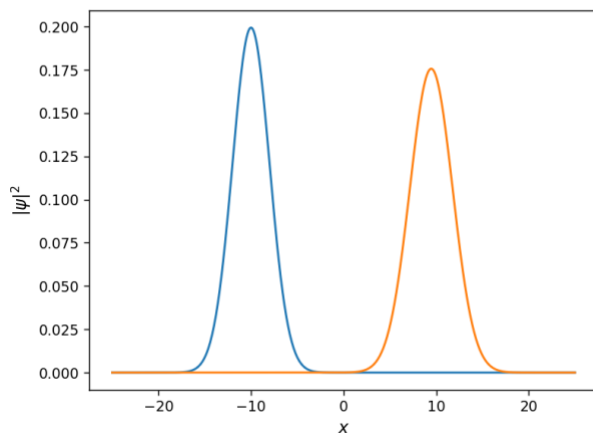
Our initial condition for solving the time-dependent Schrödinger equation is that $\psi(x,t)$ is a Gaussian wavepacket. For a particle centered at $\bar{x}$ and moving with average momentum $\bar{p}$, we will use the minimum wavepacket, that which minimally satisfies the Heisenberg uncertainty principle $\sigma_x \sigma_p = \hbar/2$. Our initial condition for the wavefunction will be

$$\psi(x, 0) = [2\pi(\delta x)^2]^{-1/4} \exp \left\{ -\frac{(x - \bar{x})^2}{4(\delta x)^2} + \frac{i\bar{p}x}{\hbar} \right\}$$

We will also use boundary conditions $\psi(x_0, t) = 0$ and $\psi(x_f, t) = 0$ (i.e., the wavefunction is zero on the boundaries for all time).

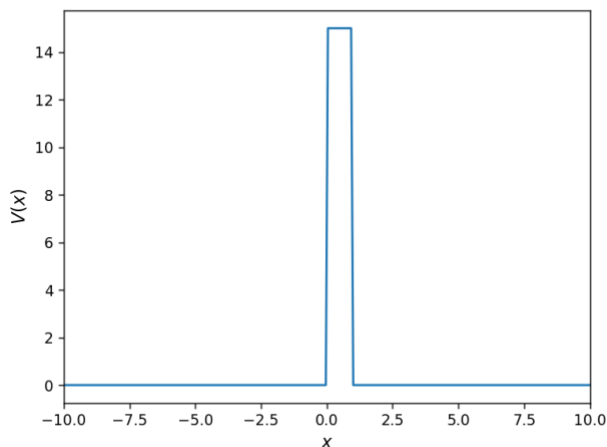
Problem 1: Motion of particle in free space

Let's first look at a wavepacket moving in free space (i.e., with $V(x,t)=0$). Using the above wavepacket as the initial condition, solve for $\psi(x,t)$ using the CN method of the Schrödinger equation above and the given boundary conditions. Start with your particle at initial mean position $\bar{x} = -10$, initial mean momentum $\bar{p} = 4$, and $\delta x = 2$, where we have used units where $\hbar = 1$ and $m = 1$. Start with $\Delta x = 0.1$ and $\Delta t = 0.001$, and calculate α (it should be < 0.5). The packet has an average momentum $\bar{p} = m\bar{v} = \hbar\bar{k}$ (with $m=1$) and should move with a constant average speed $\bar{v}$. Assuming the packet moves with constant speed, set the number of time steps so that the wavepacket travels a distance so that its final peak position is at $x=10$. Given your Δx and Δt , does the peak of your wavefunction reside near $x=10$ at the end (it should be close)? Also choose an appropriate range for your lattice to make sure your wavefunction fits into your simulation region at both the start and end of the simulation. To visualize your wavefunction, plot $|\psi(x,t)|^2$ at the beginning and end (see figure below that shows the initial wavepacket in blue and the final wavepacket in orange). As a numerical check, integrate $|\psi(x,t)|^2$ over the whole region and confirm that it is nearly equal (within numerical error) to 1.0 at both the beginning and the end. (It had better be, as no probability should be lost.)



Problem 2: Scattering off rectangular barrier

Now we will scatter off a rectangular barrier, of height V_0 , width a , and positioned from $0 < x < a$. The figure below shows a potential for $a=1$ and $V_0=15$.



Classically, a particle with energy less than the barrier height would be 100% reflected; a quantum-mechanical particle can tunnel through, so some of it gets transmitted. You will calculate how much of the wavefunction gets through, as a function of its energy. The average energy of the incident wavepacket is $\bar{E} = \frac{1}{2} \frac{\bar{p}^2}{m}$. Use wavepackets that start at $\bar{x} = -10$ but with different initial momenta $\bar{p}$ (i.e., different initial energy $\bar{E}$). For each $\bar{p}$, run your CN simulation to find the wavefunction after it has interacted with the barrier. Start with $V_0=20$ and $a=1$. Similar to what was done above, set the number of timesteps to be such that the wavepacket traveling at speed $\bar{v}$ would travel to be at final peak position $x=25$. Plot some of your final scattered wavefunctions $|\psi(x, t)|^2$.

From your wavefunction at the last timestep, the transmission probability (i.e., the likelihood that the particle tunneled through the barrier) is

$$T = \int_{x=a}^{x=L} |\psi(x, t)|^2$$

where L is the rightmost edge of your simulation domain. Calculate T over a range of initial momenta such that $0 < E < 2V_0$. Plot T vs. E/V_0 . Use $a=1$, and try $V_0=0.5, 10$, and 20 . Comment on the shape of T vs. E/V_0 .

Now fix $E = 0.5V_0$ and vary the barrier width a . Calculate T at each width a , and plot T as a function of a . What functional dependence does T seem to have with barrier width a ?

TIP: Try to simulate sufficiently long so that there is very little amplitude of the wavefunction within the barrier (i.e., $|\psi(x, t)|^2$ in the barrier is small).

DISCUSSION: Comment on the numerical advantages of CN vs., say, the FTCS method that we used in Activity Guide 7. Look up quantum tunneling on Wikipedia and discuss some of the advantages and drawbacks that tunneling has in electronics and other technological applications.